GlucoTwin AI — architecture of the predictive diabetes digital twin

Team hackers_ · IIT (ISM) Dhanbad · research prototype, not a medical device · trained on open research data (not an Indian cohort)

1. Data sources

Open research data, 187 people:
Shanghai T2DM / T1DM (with EHR), HUPA-UCM, T1D-UOM, AZT1D, BrisT1D
External test only: OhioT1DM (DUA, never trained on, never published)

Other inputs (labelled)

Synthetic demo: 120 simulated patients (separate model, marked SYNTHETIC)
User upload: own CGM CSV (Dexcom / LibreView / generic) — kept in memory only, never saved to disk
Clinician 'daily context': meal, insulin, tablets (what-if, not validated)

2. Validation & preprocessing

external_loaders.py · real_data_pipeline.py
user_upload.py (same rules for uploads)

- units: mmol/L → mg/dL, carb portions, basal U/h → U per 5 min
- glucose range 39–450 mg/dL
- 'not recorded' kept as NaN, not zero
- interpolated (future-leaking) points removed
- BrisT1D 6-h windows stitched (0 conflicts)
- > 6 h CGM gap starts a new segment
- QC report for the synthetic data

3. EHR + sensor integration

Common 15-minute grid per person (5-min and 15-min CGMs)

Dynamic stream: CGM, insulin, meals, heart rate, steps, oral-drug doses
Static stream: EHR joined per recording (age, sex, BMI, duration, HbA1c, labs, complications, drug classes)

Label: max CGM in the next 30 min / 1 h / 2 h > 180 mg/dL

4. Feature engineering

Past-only (causal) features:
- glucose level, slopes, projections, rolling stats, time above / below range
- insulin & carbs on board
- personal history: same time of day over 7 / 14 days, 72-h variability
- wearable HR / steps, time of day
- 31 EHR features

One code path for training and serving; leak tests: changing future data never changes a past feature

training

API boundary (JSON over HTTP)

5. Trained ML models

HistGradientBoosting (scikit-learn)
30 min & 1 h: v7 · 2 h: v9

Artifacts (loaded once, cached):
backend/models/twin_v7/*.joblib
backend/models/twin_v9/model_2h.joblib
Thresholds from validation only

Validation: person-grouped nested CV, locked test set (v9), OhioT1DM external, calibration, bias, verifier (297 checks)

joblib

6. FastAPI endpoints

/api/v7/patients[/{id}] /stream
/api/v7/predict/{id}?bin=
(+ optional what-if scenario)
/api/v7/patients/{id}/overview | trends
/api/v7/upload (multipart CSV)
/api/v7/model-card
/api/* synthetic demo + /api/health

Per request: raw stream up to 'now' → same feature code → 3 models (~50–100 ms)

7. React dashboard

Vite + React + Recharts (light / dark)
Patient · My data · Prediction · Live monitor · Trends · Reports · Analytics · Settings

Observed data (past, solid) is drawn apart from outcomes (replay, dashed) and from predictions (probabilities)
Modes: data · alert · theme · layout

8. Clinician-facing output

Spike probability for 30 min, 1 h, 2 h
Risk level from validated thresholds
Early warning: only while glucose is still ≤ 180 (max once per hour)
Signals the model read (explanation)
Data completeness & last reading time
Replay: what really happened next
Printable report

EXPERIMENTAL — not for clinical use

Verified held-out accuracy, 187 unseen people (7-fold person-grouped CV) · 30m: 94.7% · 60m: 91.1% · 2h: 84.9% · Type 2 only, 2 h: 89.8% · calibration (ECE) is reported separately